Closed-Loop Imitation Learning for Battery Control in Distribution Systems

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Abstract

Battery Energy Storage Systems (BESS) deployed in distribution networks are typically dispatched by a rolling-horizon linear program (LP) Energy Management System (EMS) that is re-solved every hour. While optimal, this LP becomes a computational bottleneck when embedded inside outer planning loops that require hundreds of co-simulation evaluations. We apply imitation learning—specifically Behavioral Cloning (BC) and Dataset Aggregation (DAgger)—to train a Random Forest policy that mimics the EMS expert from locally observable inputs: bus voltages, battery state of charge (SOC), and time of day. The learned policy replaces only the hourly EMS dispatch; a local droop layer handles sub-second voltage transients identically in all setups. We evaluate on the IEEE 34-bus unbalanced distribution feeder in a high-fidelity OpenDSS/HELICS co-simulation over 40-day runs across three price-noise levels ($\sigma \in \{0.004, 0.04, 0.4\}$ \$/kWh). Results show that BC matches LP revenue exactly at low noise and captures 89.8% of LP revenue at high noise while maintaining lower voltage RMSE than LP. DAgger further reduces setpoint MSE by 17% relative to BC in the hardest scenario, recovering additional economic value. Voltage quality remains essentially unchanged across all controllers, suggesting that the droop layer provides a robust safety floor that partially mitigates distribution shift.

1 Introduction

The rapid growth of distributed energy resources—particularly rooftop photovoltaics (PV) and battery energy storage systems (BESS)—is transforming electric power distribution networks [3]. BESS assets can absorb excess solar generation during midday overvoltage events and discharge during evening peak-load periods, simultaneously providing energy arbitrage revenue and voltage support. The intermittency of solar generation and the time-coupled nature of battery state-of-charge (SOC) dynamics make real-time BESS dispatch a challenging sequential decision-making problem.

The dominant approach to BESS voltage regulation is optimization-based control: at the start of each hour, a rolling-horizon EMS solves a linear program (LP) over the remaining hours of the day, computing active and reactive power setpoints $(P_h^{\mathrm{ref}}, Q_h^{\mathrm{ref}})$ that maximize energy arbitrage while keeping bus voltages within the permissible band $[V^{\min}, V^{\max}]$. Although each LP solve completes in milliseconds, the EMS is frequently embedded inside an outer planning loop that searches over hundreds of candidate BESS siting and sizing configurations, each requiring a full day-long co-simulation with 24 hourly re-solves across multiple seasonal days. The cumulative cost across all evaluations is substantial, and replacing the inner EMS with a single neural network forward pass eliminates this bottleneck entirely.

This work makes three contributions. First, we establish an empirical study of imitation learning for BESS voltage regulation in unbalanced distribution networks, providing a novel physical domain in

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which to test distribution shift theory. Second, we build a reproducible co-simulation testbed on the IEEE 34-bus feeder using OpenDSS and HELICS with multiple price-noise stress scenarios. Third, we provide a controlled quantitative comparison of BC and DAgger against the LP-EMS expert, with evaluation metrics tied directly to the theoretical predictions that distinguish the two algorithms. Among four model architectures evaluated (MLP, LSTM, LSTM with auxiliary task, and Random Forest), Random Forest consistently performed best in the BC setting—a finding that underscores the value of tabular structure in this domain.

2 Related Work

Behavioral cloning. The foundational example of BC is ALVINN [7], which trained a three-layer neural network to map raw camera images to steering commands by imitating a human driver, demonstrating that end-to-end supervised learning on expert demonstrations can produce functional control policies. ALVINN also exposed BC's central weakness: the network only performs well near states the expert visited during training. Our work applies the same imitation paradigm to BESS dispatch, with the EMS replacing the human expert and bus voltages plus SOC replacing camera pixels.

Distribution shift and DAgger. Ross, Gordon, and Bagnell [8] formalized BC's compounding error problem and proposed DAgger, proving that iterative dataset aggregation under the learner's state distribution reduces degradation from $O(T^2\epsilon)$ to $O(T\epsilon)$. Our work tests whether this advantage transfers to a physical control domain where SOC bounds and droop enforcement constrain state divergence.

Optimization-based BESS dispatch. Fortenbacher, Mathieu, and Andersson [3] present a two-stage model predictive control scheme for distributed BESS in low-voltage grids using a linearized optimal power flow formulation that jointly manages voltage regulation and battery degradation. Their LP-based MPC is the direct predecessor of the EMS expert we use as our oracle. Unlike their approach, our learned policy requires no network model or load forecast at inference time.

Learning-based voltage control. Liu and Wu [5] propose a two-stage deep reinforcement learning framework for inverter-based Volt-VAR control in active distribution networks, using adversarial offline pre-training followed by online fine-tuning. While their approach demonstrates that learning-based controllers can match optimization-based methods, RL requires careful reward shaping, slow online exploration, and safety monitoring. Our imitation learning approach avoids these issues: the EMS provides direct action supervision, BC training is purely offline, and DAgger queries the expert before any unsafe state is reached.

Learning to solve optimal power flow. Zamzam and Baker [9] train a neural network on AC optimal power flow solutions to produce near-optimal generator dispatch in milliseconds, demonstrating that learning from an optimization oracle is viable at power-systems scale. The key distinction from our work is that their setting is a single-shot regression problem, while ours is a sequential control problem where the policy's actions affect future states, making distribution shift a central concern.

ML-based battery energy management. Nakabi and Toivanen [6] apply deep reinforcement learning to battery energy management in microgrids, training a DQN-based agent to optimize charging and discharging schedules for cost minimization. Their RL agent requires tens of thousands of environment interactions and does not incorporate voltage regulation constraints. Our approach generates high-quality training data efficiently by directly imitating an EMS oracle that already satisfies all voltage constraints.

3 Problem Statement

Let the distribution feeder state at hourly boundary h be $s_h = (V_h^{\text{meas}}, \text{SOC}_{d,h})$, where $V_h^{\text{meas}} \in \mathbb{R}^{|J|}$ is the vector of measured phase voltages at all buses and $\text{SOC}_{d,h} \in [0, E_d]$ is the battery energy level. The observation available to a real-time controller is:

$$o_h = \left[V_h^{\text{meas}},\ \text{SOC}_{d,h},\ \sin(2\pi h/24),\ \cos(2\pi h/24)\right], \tag{1}$$

which includes no future load or price information. The controller outputs a power setpoint $(P_h^{\text{ref}}, Q_h^{\text{ref}})$ for the BESS unit. A sub-second droop layer then applies local corrections to enforce SOC bounds and voltage regulation between hourly re-solves.

The goal is to learn a policy $\pi_\theta : o_h \mapsto (P_h^{\text{ref}}, Q_h^{\text{ref}})$ that minimizes the expected imitation error with respect to the EMS expert π^*:

$$\min_\theta \; \mathbb{E}_{o_h \sim d_{\pi_\theta}} \Big[\big\| \pi_\theta(o_h) - \pi^*(o_h) \big\|_2^2 \Big], \tag{2}$$

where d_{π_θ} is the state distribution induced by the learner's own policy. BC minimizes this over d_{π^*} (expert's distribution), while DAgger iteratively expands the training distribution toward d_{π_θ}, reducing the compounding error bound from $O(T^2\epsilon)$ to $O(T\epsilon)$ where T is the horizon and ϵ is the per-step error [8].

4 Method

4.1 Expert Oracle: Rolling-Horizon EMS

At the start of each hour h, the EMS solves an LP over the remaining hours $\mathcal{T}_h = \{h, \ldots, 23\}$:

$$\max \sum_d \sum_{t \in \mathcal{T}_h} \pi_t P_{d,t} \Delta t - \lambda_v \sum_j (s_j^{\text{low}} + s_j^{\text{high}}) - \lambda_{vd} \sum_j v_{j,t}^{\text{dev}}, \tag{3}$$

subject to SOC dynamics with charge/discharge efficiencies η_d^{dis}, η_d^{ch}, SOC bounds $[\text{SOC}^{\min} \bar{E}_d, \, \text{SOC}^{\max} \bar{E}_d]$, and soft voltage constraints from a first-order power-flow linearization [1]:

$$\hat{V}_{j,t}^{\varphi} = V_{j,t}^{\varphi,\text{base}} + \sum_d A_{j\varphi, i_{d,t}} P_{d,t} + \sum_d B_{j\varphi, i_{d,t}} Q_{d,t}, \tag{4}$$

where A and B are voltage sensitivity coefficients computed from $Z_{LL} = Y_{LL}^{-1}$. Only the first-hour solution $(P_h^{\text{ref}}, Q_h^{\text{ref}})$ is dispatched; the LP is re-solved at $h+1$ from the measured SOC.

4.2 Droop Layer

Between hourly re-solves, a Volt-VAR droop adjusts reactive power as a piecewise linear function of local bus voltage per IEEE 1547-2018, and a SOC-aware Volt-Watt droop applies active power corrections:

$$P_d^f(t) = \min\left(\bar{P}_d, \; \frac{(\text{SOC}(t) - \text{SOC}^{\min}) \bar{E}_d \, \eta^{\text{dis}}}{\Delta t_c / 3600} \right), \tag{5}$$

preventing SOC violations regardless of the hourly setpoint. This layer is *identical* in the LP-EMS, BC, and DAgger setups; the only difference between setups is what produces the hourly reference $(P_h^{\text{ref}}, Q_h^{\text{ref}})$.

4.3 Co-simulation Environment

We use OpenDSS [2] coupled with HELICS [4] on the IEEE 34-bus unbalanced three-phase radial test feeder. Figure 1 illustrates the co-simulation architecture. HELICS coordinates five federates: the *Grid Federate* (OpenDSS, runs every 5 min), the *EMS/Policy Federate* (LP or learned policy, runs every 60 min), the *Droop Control Federate* (fast voltage regulation, every 5 min), a *Random Price Generator*, and a *Logger Federate*. The measured SOC from OpenDSS is fed back to the dispatch federate at every hourly boundary.

4.4 Behavior Cloning: Policy Training

The network π_θ receives observation o_h (Eq. 1) and outputs $(P_h^{\text{ref}}, Q_h^{\text{ref}})$. We evaluated four model architectures: MLP, LSTM, LSTM with auxiliary task (predicting later-horizon powers in addition to the immediate setpoint), and Random Forest (RF). All models are trained by minimizing mean-squared error against EMS labels on a dataset collected from EMS rollouts spanning four seasonal load profiles (winter peak, spring shoulder, summer peak, fall shoulder) and three PV penetration levels (0%, 50%, 100% of feeder peak load).

The time embedding $(sin(2\pi h/24), \cos(2\pi h/24))$ ensures that midnight and 11 pm are represented as adjacent points on the unit circle rather than numerically distant integers, allowing the policy to distinguish time-dependent dispatch strategies without accessing future information.

[Figure: Co-simulation Architecture]

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HELICS Broker $\leftrightarrow$ Grid Federate (OpenDSS, 5-min) | EMS/Policy Federate (60-min) | Droop Federate (5-min) | Logger Federate.

Signals: P_{ref}, SOC, P_{act}, Voltage.

Figure 1: Co-simulation architecture. Five HELICS federates coordinate on the IEEE 34-bus feeder. The EMS/Policy federate is the only component that differs between LP-EMS, BC, and DAgger setups.

4.5 DAgger

Starting from the BC dataset, DAgger iterates: (1) roll out the current policy π_θ^{i-1} in the co-simulation; (2) query the EMS for the optimal action a_h^* at every visited state; (3) aggregate the new transitions into the training set; (4) retrain π_θ^i. Price-stress rollouts are explicitly included in DAgger iterations so the expert is queried at exactly the out-of-distribution states where BC is hypothesized to fail.

5 Experimental Design

5.1 Domain and Data

All experiments use the IEEE 34-bus three-phase unbalanced radial distribution feeder in OpenDSS with HELICS co-simulation. A single BESS unit is placed at a high-voltage-sensitivity bus. We test three levels of price-forecast noise: $\sigma \in \{0.004, 0.04, 0.4\}$ $/kWh, where higher noise creates greater out-of-distribution pressure on BC. Each run spans 40 simulated days at 5-minute resolution (960 hours, 11,520 timesteps).

5.2 Hypotheses

H1: BC produces a functional dispatch policy at low noise ($\sigma = 0.004$), matching LP revenue and voltage quality. **H2:** Higher noise ($\sigma = 0.04, 0.4$) exposes distribution shift, increasing the setpoint MSE between BC and LP. **H3:** DAgger trained on high-noise rollouts recovers BC's degradation, achieving lower setpoint MSE and higher revenue than BC alone.

5.3 Evaluation Metrics

The primary metrics are: (i) *Revenue proxy* ($\sum_t \pi_t P_{\mathrm{act},t} \Delta t$, in $), measuring economic performance; (ii) *Voltage RMSE* ($\sqrt{\mathbb{E}[(V_{\mathrm{pu}} - 1)^2]}$, in pu), measuring safety/quality; and (iii) *Setpoint MSE* ($\mathbb{E}[\|P_{\mathrm{BC}}^{\mathrm{ref}} - P_{\mathrm{LP}}^{\mathrm{ref}}\|^2]$, in kW^2), measuring imitation accuracy. LP serves as the economic ceiling across all conditions.

6 Results and Discussion

6.1 Environment Validation

Before training any policies, we validated the co-simulation environment itself. The HELICS framework successfully coordinated all five federates on the IEEE 34-bus feeder, producing stable 48-hour trajectories with consistent SOC cycling, realistic voltage profiles, and economically sensible dispatch patterns. The LP-EMS federate converged at every hourly boundary, and the droop layer maintained voltages within acceptable bounds during 5-minute sub-steps. This validation was a non-trivial engineering milestone: the multi-federate, time-coupled, physics-constrained setup required careful synchronization of HELICS grants and careful handling of boundary conditions between the optimization and simulation layers.

[SOC trajectory]
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[Power setpoints]
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(a) SOC over 48 h (b) Bus voltage (pu) (c) P_{ref} vs P_{act}

Figure 2: LP-EMS closed-loop validation on IEEE 34-bus. 48-h run, 5-min resolution, price noise $\sigma = 0.004$ $/kWh (seed 32). The SOC trajectory remains within bounds, voltage stays near 1 pu, and the droop layer smoothly tracks the hourly setpoints.

6.2 Behavior Cloning Results

Table 1 summarizes all experimental runs. At low noise ($\sigma = 0.004$), BC achieved revenue identical to LP ($6,272) with the same voltage RMSE (0.02304 pu), confirming **H1**: BC produces a functionally equivalent policy on in-distribution scenarios. This result demonstrates that the Random Forest can perfectly replicate the LP's dispatch decisions when price uncertainty is minimal and the expert's visited states span the operating space.

At medium noise ($\sigma = 0.04$), LP revenue increased to $13,631 due to higher price volatility, while BC captured $12,291 (90.2%), with a setpoint MSE of 30,623 kW^2. At high noise ($\sigma = 0.4$), LP revenue grew dramatically to $120,712, while BC captured $108,440 (89.8%), with setpoint MSE rising to 39,168 kW^2. The increasing revenue gap and MSE with noise confirm **H2**: distribution shift degrades BC under higher out-of-distribution pressure.

Notably, BC's voltage RMSE was *lower* than LP's at both medium and high noise (0.02320 vs. 0.02496 and 0.02389 vs. 0.02523), suggesting that the BC policy's smoother dispatch—even when economically suboptimal—causes less voltage fluctuation than the LP's aggressive arbitrage strategies.

Table 1: Summary of all experimental results. Revenue proxy is $\sum_t \pi_t P_{\mathrm{act},t} \Delta t$ [$]; Voltage RMSE is $\sqrt{\mathbb{E}[(V_{\mathrm{pu}} - 1)^2]}$ [pu]; Setpoint MSE is $\mathbb{E}[\|P_{\mathrm{BC}}^{\mathrm{ref}} - P_{\mathrm{LP}}^{\mathrm{ref}}\|^2]$ [kW^2]. Runs span 40 simulated days (960 h) at 5-min resolution.

Controller	Noise σ	Revenue [$]	V-RMSE [pu]	MSE vs. LP [kW^2]
LP-EMS	0.004	6,272	0.02304	—
BC (RF)	0.004	6,272	0.02304	—
LP-EMS	0.04	13,631	0.02496	—
BC (RF)	0.04	12,291	0.02320	30,623
LP-EMS	0.4	120,712	0.02523	—
BC (RF)	0.4	108,440	0.02389	39,168
DAgger	0.4	109,820	0.02391	32,514

6.3 DAgger vs. BC

DAgger was evaluated at the highest noise level ($\sigma = 0.4$), the setting where BC showed the most degradation. DAgger improved over BC on both imitation accuracy and economic performance: setpoint MSE dropped from 39,168 to 32,514 kW^2 (a 17% reduction), and revenue increased from $108,440 to $109,820. Voltage RMSE remained essentially unchanged (0.02391 vs. 0.02389), confirming that the droop layer provides a robust voltage safety floor independent of the hourly dispatch quality. These results confirm **H3**: DAgger recovers part of BC's degradation on the hardest scenario.

6.4 Model Architecture Comparison

Among the four architectures evaluated in the BC setting—MLP, LSTM, LSTM with auxiliary task (predicting future-horizon powers), and Random Forest—the Random Forest consistently performed best. This finding is somewhat surprising given the sequential nature of the problem, but it is consistent with the highly structured, tabular character of the observations: at each hourly boundary, the EMS dispatch depends heavily on the current SOC and time of day in a non-linear but non-recurrent way, which RF captures well via decision tree ensembles. The LSTM with auxiliary task showed the most stable long-run behavior—it did not collapse under distribution shift even in 40-day runs, with SOC trajectories remaining dynamic rather than becoming stuck at a boundary.

(a) BC (RF), $\sigma = 0.4$, 48-h run

(b) BC (RF), 40-day SOC trajectory

Figure 3: BC Random Forest closed-loop results. Left: 48-h trajectory under $\sigma = 0.4$ price noise (seed 11). Right: 40-day SOC trajectory showing that BC does not collapse even in extended runs—SOC remains dynamic rather than saturating at a boundary.

6.5 Long-Run Stability

Despite being trained on shorter horizons, none of the BC models collapsed in 40-day closed-loop runs. SOC trajectories remained dynamic and bounded, suggesting that the droop layer's hard SOC enforcement prevents the runaway divergence that standard distribution shift theory predicts for unbounded domains. This partially mitigates H2: while setpoint MSE increases with noise, the physical bounds limit how severely the learned policy can degrade in terms of actual power system behavior.

7 Limitations and Future Work

Several limitations constrain the current study. First, we use a single BESS unit; multi-battery coordination introduces coupling between agents that imitation learning would need to address jointly. Second, DAgger was run for only one iteration due to computational cost; full convergence would likely close more of the gap to LP. Third, the EMS uses a linearized power flow model [1], which introduces modeling error that the learned policy inherits. Fourth, all experiments use a single synthetic load profile; real-world seasonal diversity and weather uncertainty are not fully captured.

Future work could explore: (i) running DAgger for multiple iterations to study convergence; (ii) replacing the RF with a graph neural network that explicitly encodes the feeder topology; (iii) testing on multi-BESS feeders and larger networks such as the IEEE 123-bus system; (iv) incorporating uncertainty quantification so the policy can flag when it is operating far from training states; and (v) integrating the learned policy into the outer siting-and-sizing planning loop to measure the actual speedup over repeated LP solves.

8 Conclusion

We presented a closed-loop imitation learning framework for BESS dispatch in unbalanced distribution networks, using a high-fidelity OpenDSS/HELICS co-simulation as the training and evaluation environment. A Random Forest policy trained by BC matched LP revenue exactly at low noise and retained 89.8% of LP revenue at high noise, while achieving lower voltage RMSE than LP in all out-of-distribution conditions. DAgger reduced setpoint MSE by 17% and recovered additional revenue in the hardest scenario. The key finding is that the physical droop layer provides a robust safety floor that prevents SOC runaway and limits voltage violations even when the learned policy drifts from the expert's distribution—a meaningful mitigant of distribution shift in safety-critical physical systems. These results support the viability of replacing online LP solves with learned policies in planning-loop applications where inference speed is critical.

Code: `https://github.com/ali-tasavvori/advanced-AI-project`

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